PHS 220 · PHYSIOLOGY STUDY GUIDE

# Endocrine Physiology

Principles, Thyroid, Calcium & Parathyroid, Pancreas & Adrenal

- General Principles of Endocrine Physiology
- Thyroid Physiology & Disorders
- Parathyroid, Calcium & Bone Homeostasis
- Pancreatic Endocrine Physiology
- Adrenal Physiology & Disorders

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# Table of Contents

# 1 GENERAL PRINCIPLES OF ENDOCRINE PHYSIOLOGY

Endocrine physiology is fundamentally the study of how the body maintains homeostasis, the stable internal environment that lets cells and organs work well despite a constantly changing outside world. The endocrine system does this by releasing chemical messengers called hormones into the bloodstream. These hormones travel to distant target tissues and produce specific biological responses. The word hormone comes from the Greek hormaein, meaning to set in motion. Hormones are secreted in tiny amounts (picomolar to nanomolar) and act only on cells that carry the matching receptor. Their reach is enormous: metabolism, growth, reproduction, fluid and electrolyte balance, and the response to stress.

**CORE PRINCIPLE**

The endocrine system never works alone. It is tightly integrated with the nervous and immune systems. Neuroendocrine links (such as the hypothalamo-pituitary axis) and immunoendocrine cross-talk (such as cortisol suppressing inflammation) sit at the centre of both normal physiology and disease.

## 1.1 Types of chemical signalling

| Signalling type | Route | Distance | Example |
|---|---|---|---|
| **Endocrine** | Bloodstream | Long range (systemic) | Insulin (pancreas to liver) |
| **Paracrine** | Interstitial fluid | Local, neighbouring cells | Somatostatin (delta to beta cells) |
| **Autocrine** | Same cell | Cell acts on itself | Growth factors in tumour cells |
| **Neurocrine** | Synapse / neuroendocrine | Targeted along an axon | ADH (hypothalamus to kidney) |
| **Juxtacrine** | Direct cell contact | Adjacent cells only | Notch-Delta signalling |

**Major hormonal contributors to homeostasis**

Thyroid hormone controls basal metabolism in most tissues. Cortisol governs energy metabolism and is permissive for other hormones. Aldosterone regulates plasma volume through electrolytes. Vasopressin (ADH) regulates plasma osmolality through water excretion. PTH regulates calcium and phosphate. Insulin regulates plasma glucose.

## 1.2 Classes of Hormones

Hormones are chemically diverse, and their chemistry decides how they are made, stored, carried in blood, and how they signal. There are three principal classes.

### Peptide and protein hormones

- Made on ribosomes as pre-prohormones, then prohormones, then active hormone (for example insulin from proinsulin).
- Water soluble, so they travel freely in plasma with a short half-life (minutes).
- Cannot cross the cell membrane, so they act on surface receptors (GPCRs, receptor tyrosine kinases, cytokine receptors).
- Signal through second messengers such as cAMP, IP3/DAG, and calcium.
- Examples: insulin, glucagon, GH, ACTH, PTH, GnRH, LH, FSH, oxytocin, ADH.

### Steroid hormones

- Derived from cholesterol in the adrenal cortex, gonads, and placenta.
- Lipid soluble, so they cross membranes freely and have a long half-life (hours).
- Carried in plasma bound to proteins (CBG, SHBG, albumin); only the free fraction is active.

- Act on intracellular (nuclear) receptors; the hormone-receptor complex binds hormone response elements on DNA to control transcription.
- Effects are slow but long lasting. Examples: cortisol, aldosterone, testosterone, oestrogen, progesterone, calcitriol (vitamin D).

### Amine (tyrosine-derived) hormones

| Hormone | Gland | Solubility | Receptor | Half-life |
|---|---|---|---|---|
| **Adrenaline / noradrenaline** | Adrenal medulla | Water soluble | Cell surface (alpha/beta) | Seconds to minutes |
| **Dopamine** | Hypothalamus / adrenal | Water soluble | Cell surface (D receptors) | Minutes |
| **Thyroid hormones (T3/T4)** | Thyroid | Lipid soluble | Nuclear (TR) | T4 ~7 days, T3 ~1 day |

## 1.3 Hormone Receptors

A receptor is a protein that binds its hormone (the ligand) with high affinity and selectivity, and upon binding triggers a response in the target cell. The type and number of receptors on a cell decide how sensitive it is to a given hormone.

### Cell-surface (membrane) receptors

| Receptor class | Coupling | Second messenger | Hormone examples |
|---|---|---|---|
| **GPCR (Gs)** | Adenylyl cyclase up | cAMP to PKA | Glucagon, TSH, ACTH, LH, FSH, PTH, adrenaline (beta) |
| **GPCR (Gi)** | Adenylyl cyclase down | cAMP down | Somatostatin, adrenaline (alpha-2) |
| **GPCR (Gq)** | Phospholipase C | IP3 + DAG to calcium, PKC | GnRH, TRH, oxytocin, adrenaline (alpha-1) |
| **Receptor tyrosine kinase** | Autophosphorylation | RAS-MAPK, PI3K-Akt | Insulin, IGF-1, EGF, PDGF |
| **JAK-STAT** | JAK activation | STAT phosphorylation | GH, prolactin, cytokines |
| **Guanylyl cyclase** | Direct GC activation | cGMP to PKG | ANP, BNP |

### Intracellular receptors

Lipophilic hormones cross the membrane and bind intracellular receptors that act as ligand-activated transcription factors. Once bound, they dimerise, move to the nucleus, and bind hormone response elements in the promoters of target genes to switch transcription on or off.

> **Receptor regulation**
>
> Down-regulation: prolonged hormone exposure lowers receptor number (internalisation and degradation). This underlies tachyphylaxis, for example prolonged GnRH agonist therapy paradoxically suppressing LH and FSH.
>
> Up-regulation: low hormone levels or opposing hormones raise receptor number and target sensitivity, for example oestrogen up-regulating uterine progesterone receptors.

## 1.4 Principles of Feedback Control

Feedback loops keep hormone concentrations within a narrow physiological range. The hypothalamo-pituitary-target organ axes are the classic example of multi-level feedback.

### Negative feedback (most common)

The product of a hormonal cascade feeds back to inhibit the components upstream, preventing excess hormone action. This dominates the HPT (thyroid), HPA (adrenal), and HPG (gonadal) axes. In the HPA axis, CRH drives ACTH, which drives cortisol; rising cortisol then suppresses both CRH and ACTH.

### Positive feedback

Here the product amplifies the upstream signal to drive a cascade to completion. It is rarer but critical at key moments. The best endocrine example is the mid-cycle LH surge: once oestrogen rises above a threshold it switches from negative to positive feedback on the pituitary, triggering the explosive LH surge that causes ovulation.

## 1.5 Pulsatile Secretion & Circadian Rhythms

Many hormones are released in discrete pulses rather than continuously. Pulsatility prevents receptor down-regulation, allows fine amplitude control, and creates windows of target sensitivity. The master clock is the suprachiasmatic nucleus of the hypothalamus, entrained by light and dark, which drives rhythmic secretion.

| Hormone | Rhythm | Peak | Clinical relevance |
|---|---|---|---|
| **Cortisol** | Circadian (~24 h) | Early morning (6-8 AM) | Rhythm lost in Cushing syndrome; measure AM cortisol |
| **GH** | Pulsatile, nocturnal | Soon after sleep onset | Continuous GH is less anabolic |
| **LH/FSH** | Pulsatile (GnRH-driven) | Varies across cycle | Continuous GnRH down-regulates receptors |
| **TSH** | Mild circadian | Late evening / night | Blunted pulse in hypothyroidism |
| **Melatonin** | Circadian | During darkness | Sleep-wake and seasonal reproduction |
| **Insulin** | Prandial + basal pulses | Post-meal | Pulsatile delivery to liver more effective |

## 1.6 Endocrine Organs

### Traditional endocrine organs

| Organ | Location | Key hormones | Primary function |
|---|---|---|---|
| **Hypothalamus** | Floor of diencephalon | TRH, CRH, GnRH, GHRH, somatostatin, dopamine | Master regulator; integrates neural and endocrine signals |
| **Anterior pituitary** | Sella turcica | TSH, ACTH, GH, FSH, LH, prolactin | Trophic hormones for peripheral glands |
| **Posterior pituitary** | Sella turcica | ADH, oxytocin | Releases hypothalamic hormones |
| **Thyroid** | Anterior neck | T3, T4, calcitonin | Basal metabolic rate; calcium |
| **Parathyroids (4)** | Posterior thyroid | PTH | Calcium and phosphate homeostasis |
| **Adrenal cortex** | Retroperitoneal | Cortisol, aldosterone, DHEA | Stress, fluid balance, sex steroids |
| **Adrenal medulla** | Core of adrenal | Adrenaline, noradrenaline | Fight-or-flight response |
| **Pancreatic islets** | Within exocrine pancreas | Insulin, glucagon, somatostatin | Glucose homeostasis |
| **Gonads** | Pelvis / scrotum | Oestrogen, progesterone / testosterone | Reproduction; secondary sex characters |
| **Pineal** | Posterior diencephalon | Melatonin | Circadian and seasonal rhythm |

### Non-traditional endocrine organs

| Organ | Hormones | Function |
|---|---|---|
| **Heart (atria)** | ANP, BNP | Natriuresis; lower BP and volume |
| **Kidney** | EPO, renin, calcitriol | RBC production; RAAS; calcium absorption |
| **Liver** | IGF-1, angiotensinogen, thrombopoietin | Growth; BP; platelet production |
| **Adipose** | Leptin, adiponectin, resistin | Satiety; insulin sensitivity; energy balance |
| **GI tract** | GIP, GLP-1, CCK, ghrelin, secretin | Digestion; incretin effect; appetite |
| **Skin** | Cholecalciferol precursor | UV-driven vitamin D synthesis |
| **Placenta** | hCG, hPL, oestrogen, progesterone | Pregnancy maintenance; fetal development |

## 1.7 The Hypothalamo-Pituitary Axis

The hypothalamo-pituitary axis is the supreme command centre of the endocrine system. The hypothalamus receives input from the cortex, limbic system, and peripheral hormones, then translates it into coordinated output through the pituitary.

### The hypothalamus

The hypothalamus secretes releasing and inhibiting hormones into the hypothalamo-hypophyseal portal system, a dedicated vascular link to the anterior pituitary. This portal route delivers hormones to the pituitary at concentrations up to a thousand times higher than in the general circulation.

| Hypothalamic hormone | Action on anterior pituitary |
|---|---|
| **TRH** | Stimulates TSH and prolactin release |
| **CRH** | Stimulates ACTH release |
| **GnRH** | Stimulates LH and FSH (pulsatile delivery required) |
| **GHRH** | Stimulates GH release |
| **Somatostatin (GHIH)** | Inhibits GH and TSH |
| **Dopamine (PIH)** | Tonically inhibits prolactin |

### Anterior pituitary (adenohypophysis)

| Cell type | Hormone | Target / effect |
|---|---|---|
| **Somatotrophs (50%)** | GH | Liver to IGF-1; growth, lipolysis, gluconeogenesis |
| **Corticotrophs (20%)** | ACTH | Adrenal cortex to cortisol and androgens |
| **Thyrotrophs (5%)** | TSH | Thyroid to T3/T4 |
| **Gonadotrophs (10%)** | FSH and LH | Gonads to sex steroids, gametogenesis, ovulation |
| **Lactotrophs (15%)** | Prolactin | Breast milk production; inhibits GnRH axis |

### Posterior pituitary (neurohypophysis)

The posterior pituitary makes no hormones of its own. It stores and releases ADH and oxytocin, which are made in the supraoptic and paraventricular nuclei of the hypothalamus and travel down axons for release into the circulation.

| Hormone | Synthesis site | Main stimuli | Key actions |
|---|---|---|---|
| **ADH (vasopressin)** | Supraoptic nucleus | Hyperosmolality, hypovolaemia, pain | V2: water reabsorption in collecting duct; V1: vasoconstriction |
| **Oxytocin** | Paraventricular nucleus | Cervical dilation, suckling | Uterine contraction; milk let-down; bonding |

### 1.8 Principles Governing Disease States

Endocrine disease falls into hormone excess, hormone deficiency, or resistance to hormone action. Identifying the axis affected and the level of the lesion is the key to diagnosis.

| Category | Definition | Lab pattern | Example |
|---|---|---|---|
| **Primary hypofunction** | Target gland fails | Low end-hormone, HIGH trophic hormone | Addison disease (low cortisol, high ACTH) |
| **Secondary hypofunction** | Pituitary/hypothalamus fails | Low end-hormone, low trophic hormone | Secondary hypothyroidism (low T4, low TSH) |
| **Primary hyperfunction** | Autonomous gland over-secretion | HIGH end-hormone, low trophic hormone | Conn syndrome (high aldosterone, low renin) |
| **Secondary hyperfunction** | Excess trophic drive | HIGH end-hormone, HIGH trophic hormone | Cushing disease (high cortisol, high ACTH) |
| **Hormone resistance** | Target unresponsive | HIGH hormone, features of deficiency | Type 2 diabetes (high insulin, hyperglycaemia) |
| **Ectopic secretion** | Non-endocrine tumour secretes hormone | HIGH hormone, source not the gland | Ectopic ACTH (small cell lung cancer) |

> **CLINICAL PEARL**
>
> The ratio of trophic hormone to end-hormone localises the lesion in any endocrine axis. Primary gland failure raises the trophic hormone (loss of negative feedback). Secondary or tertiary failure lowers it. This applies equally to the HPT, HPA, and HPG axes.

## 2 THYROID PHYSIOLOGY & DISORDERS

The thyroid is a highly vascular, butterfly-shaped gland in the anterior neck, one of the largest endocrine glands at 20 to 30 g in adults. Its function matters to every cell in the body, because thyroid hormones set the pace of basal metabolism.

### 2.1 Gross anatomy

| Feature | Description |
|---|---|
| **Shape** | Two lateral lobes joined by an isthmus; a pyramidal lobe in about half of people |
| **Position** | Anterior to the 2nd to 4th tracheal rings, deep to the strap muscles |
| **Relations** | Posterolateral carotid sheath; posterior parathyroids and recurrent laryngeal nerves |
| **Arterial supply** | Superior thyroid artery (from ECA) and inferior thyroid artery (from thyrocervical trunk) |
| **Nerve supply** | Vasomotor only from the superior cervical ganglion; secretion is TSH-driven |

### 2.2 Microscopic structure

The functional unit is the follicle, a sphere lined by follicular cells (thyrocytes) around a lumen of colloid (mainly thyroglobulin). The gland holds around 3 million follicles.

| Cell / structure | Location | Function |
|---|---|---|
| **Follicular cells (thyrocytes)** | Follicle wall | Synthesise and secrete T3/T4; trap iodide; make thyroglobulin |
| **Colloid** | Follicle lumen | Stores thyroglobulin-bound hormone (2 to 3 months supply) |
| **Parafollicular cells (C-cells)** | Interfollicular stroma | Secrete calcitonin in response to hypercalcaemia |

**HISTOLOGY NOTE**

A follicle's appearance reflects its activity. An inactive follicle has abundant colloid and flat squamous thyrocytes. An active follicle has little colloid and tall cuboidal-to-columnar thyrocytes, reflecting TSH-driven up-regulation of secretory machinery.

## 2.3 Thyroid Hormone Production

Hormone synthesis is a unique multi-step process that needs dietary iodine, about 150 micrograms daily in adults and 200 in pregnancy. It occurs at the interface of the thyrocyte apical membrane and the colloid.

- Step 1, iodide trapping: the sodium-iodide symporter (NIS) on the basolateral membrane actively pumps iodide into the cell, concentrating it 30 to 250 times. NIS is the target of radioiodine therapy.
- Step 2, efflux into colloid: pendrin moves iodide across the apical membrane. Pendrin mutations cause Pendred syndrome (congenital hypothyroidism plus deafness).
- Step 3, thyroglobulin synthesis: a large glycoprotein made by the rough ER and exocytosed into colloid; it is the scaffold on which hormone is built.
- Step 4, oxidation: thyroid peroxidase (TPO) with hydrogen peroxide oxidises iodide to reactive iodine. Blocked by propylthiouracil and carbimazole.
- Step 5, organification: iodine is added to tyrosine residues on thyroglobulin, forming MIT and DIT (also TPO-catalysed).
- Step 6, coupling: MIT + DIT gives T3; DIT + DIT gives T4, still bound within thyroglobulin.
- Step 7, storage: hormone stays stored on thyroglobulin for 2 to 3 months, which is why antithyroid drugs take weeks to work.
- Step 8, secretion: TSH triggers colloid pinocytosis; lysosomal cathepsins cleave T3/T4 for secretion, and leftover MIT/DIT are deiodinated to recycle iodide.

**DRUG TARGET SUMMARY**

PTU and carbimazole/methimazole block TPO, inhibiting organification and coupling; PTU also blocks peripheral T4 to T3 conversion (preferred in thyroid storm and first-trimester pregnancy).

Radioiodine (I-131) is taken up by NIS and destroys follicular cells (Graves disease, differentiated thyroid cancer).

Lithium blocks thyroglobulin proteolysis and hormone release.

Iodine excess transiently inhibits synthesis (the protective Wolff-Chaikoff effect).

## 2.4 Thyroid Hormones: Chemistry & Transport

| Property | T4 (thyroxine) | T3 (triiodothyronine) |
|---|---|---|
| **Iodine atoms** | 4 | 3 |
| **Secretion ratio** | ~80% | ~20% |
| **Potency** | 1x (reference) | 3 to 4x more potent |
| **Plasma half-life** | ~7 days | ~1 day |
| **Free fraction (active)** | ~0.03% | ~0.3% |
| **Role** | Main circulating form; prohormone for T3 | Principal active form at the receptor |

Most hormone travels bound to plasma proteins: thyroxine-binding globulin (TBG, the main high-affinity carrier), transthyretin, and albumin (low affinity, high capacity). Only the free fraction is active.

> **CLINICAL PEARL: total versus free hormone**
> Changes in TBG alter total hormone levels but not the free fraction or clinical status. Pregnancy and oestrogen raise TBG, so total T4 rises while free T4 stays normal and the patient is euthyroid. Interpret total T3/T4 against TBG, or measure free T4 and free T3 directly.

## 2.5 Physiological Effects of Thyroid Hormones

Thyroid hormones act on nearly every organ, and their overarching effect is to raise the basal metabolic rate. Total absence drops BMR 40 to 50% below normal; extreme excess raises it 60 to 100% above normal.

| System | Effect | Clinical parallel |
|---|---|---|
| **Metabolism (BMR)** | Up O2 use, heat, glucose absorption, gluconeogenesis, lipolysis; down cholesterol | Hyper: weight loss, heat intolerance; Hypo: weight gain, cold intolerance, high cholesterol |
| **Cardiovascular** | Up HR, CO, contractility; down SVR; sensitises to catecholamines | Hyper: palpitations, AF; Hypo: bradycardia, pericardial effusion |
| **CNS / development** | Essential for brain development; maintains mentation and mood | Congenital deficiency: cretinism; adults: hypo slows cognition, hyper causes anxiety and tremor |
| **GI** | Up motility, appetite, gastric acid | Hyper: diarrhoea; Hypo: constipation |
| **Reproductive** | Needed for normal sex hormone metabolism | Hypo: menorrhagia, anovulation, infertility |
| **Growth** | Synergises with GH; bone maturation | Hypo children: short stature, delayed bone age |

## 2.6 Regulation of Thyroid Hormone Secretion

Secretion is controlled by a classic negative feedback loop. The hypothalamus releases TRH, which drives pituitary TSH. TSH binds thyrocyte receptors (Gs-coupled, cAMP-PKA), stimulating every step of synthesis and secretion and promoting thyroid growth. Rising T3/T4 suppress both TRH and TSH, with T3 generated locally in thyrotrophs being the main feedback signal. Cold exposure raises TRH and thus thermogenesis.

| Iodine state | Effect on thyroid |
|---|---|
| **Iodine deficiency** | Hypertrophy (goitre); preferential T3; low T4; high TSH |
| **Acute iodine load (Wolff-Chaikoff)** | Transient inhibition of organification; normal gland escapes in 10 to 14 days |
| **Persistent high iodine** | May fail to escape in Hashimoto (hypothyroidism), or trigger Jod-Basedow hyperthyroidism in MNG |
| **Lugol iodine (pre-op)** | Reduces vascularity and hormone release before thyroidectomy |

## 2.7 Mechanism of Action

T3, the active form, crosses the membrane and binds thyroid hormone receptors (TRalpha and TRbeta) of the nuclear receptor family. TR forms a heterodimer with the retinoid X receptor and binds thyroid hormone response elements. Without T3 the complex recruits co-repressors and silences transcription; with T3 it recruits co-activators and switches transcription on. This genomic action is slow (hours to days) but sustained, building proteins such as Na/K-ATPase subunits, myosin heavy chains, and GLUT4. Rapid non-genomic actions at the membrane and mitochondria (via PI3K, MAPK, PKA, PKC) explain the quick cardiovascular effects.

## 2.8 Disorders of the Thyroid Gland

### Hyperthyroidism (thyrotoxicosis)

| Feature | Graves disease | Toxic MNG | Toxic adenoma |
|---|---|---|---|
| **Mechanism** | TSI activate the TSH receptor | Multiple autonomous nodules | Single autonomous nodule |
| **Specific features** | Ophthalmopathy, pretibial myxoedema, acropachy | Older patients; iodine-deficient areas | Solitary hot nodule on scan |
| **TSH** | Suppressed | Suppressed | Suppressed |
| **Scan (I-123)** | Diffuse uptake | Patchy uptake | Solitary hot nodule; rest cold |
| **Treatment** | Antithyroid drugs, radioiodine, surgery | Radioiodine or surgery | Radioiodine or surgery |

**Symptoms of hyperthyroidism**

Metabolic: weight loss despite increased appetite, heat intolerance, sweating. Cardiovascular: palpitations, tachycardia, atrial fibrillation, wide pulse pressure. Neurological: anxiety, fine tremor, insomnia, hyperreflexia. Musculoskeletal: proximal myopathy, osteoporosis. GI: diarrhoea. Labs: high T3/T4, suppressed TSH, positive TSI in Graves.

### Hypothyroidism

| Type | Cause | Key features | Lab pattern |
|---|---|---|---|
| **Primary** | Hashimoto (anti-TPO), post-radioiodine, post-surgery, iodine deficiency | Fatigue, weight gain, cold intolerance, constipation, dry skin, slow reflexes, bradycardia | HIGH TSH, low FT4, anti-TPO in Hashimoto |
| **Secondary** | Pituitary failure (Sheehan, adenoma, irradiation) | Milder; other pituitary deficits | Low TSH, low FT4 |
| **Subclinical** | Borderline primary failure | Often asymptomatic | High TSH (4 to 10), normal FT4 |
| **Congenital (cretinism)** | Agenesis, synthesis defect, maternal iodine deficiency | Irreversible intellectual disability if untreated | High TSH, low T4 on neonatal screen |

**EMERGENCY: thyroid storm**

A life-threatening flare of hyperthyroidism, usually triggered by stress (surgery, infection, trauma, labour). Features: fever above 38.5 C, severe tachycardia (often AF), agitation or psychosis, diarrhoea, vomiting, and high-output cardiac failure. Mortality over 20% untreated.

Treatment: PTU (blocks synthesis and conversion), then Lugol iodine, beta-blockers (propranolol), hydrocortisone (prevents adrenal crisis), plus supportive care.

# 3 PARATHYROID, CALCIUM & BONE HOMEOSTASIS

The four parathyroid glands are tiny (about 6 by 4 by 2 mm, 40 mg each) and sit on the posterior surface of the thyroid, two superior and two inferior. Despite their size they are vital: complete removal causes fatal hypocalcaemic tetany within days.

| Feature | Description |
|---|---|
| **Number** | Usually 4 (occasionally 3 to 6); ectopic supernumerary glands possible |
| **Blood supply** | Mainly the inferior thyroid arteries; preserve during thyroidectomy |
| **Embryology** | Superior pair from 4th pouch; inferior pair from 3rd pouch (with thymus) |
| **Chief cells** | Main cells; synthesise and secrete PTH |
| **Oxyphil cells** | Function unclear; increase after puberty |

### 3.1 Parathyroid hormone (PTH)

| Property | Detail |
|---|---|
| **Chemical class** | Single-chain peptide of 84 amino acids |
| **Synthesis** | Pre-pro-PTH to pro-PTH to PTH, cleaved in ER and Golgi |
| **Active fragment** | N-terminal PTH(1-34) carries full activity |
| **Half-life** | 2 to 4 minutes; cleared by liver and kidney |
| **Receptor** | PTH1R, a class B GPCR; mainly Gs to cAMP, also Gq |

**PTH secretion: key values**

Primary stimulus is hypocalcaemia (ionised calcium below about 1.1 mmol/L). The chief cell senses ionised calcium through the calcium-sensing receptor (CaSR), a surface GPCR. When ionised calcium falls, CaSR activity falls and PTH secretion rises (an inverse relationship).

Other stimuli: low magnesium, high phosphate. Inhibitors: high calcium, high magnesium, and calcitriol (long-loop negative feedback).

## 3.2 Importance of Calcium

Calcium is the most abundant mineral in the body (about 1 kg, 99% in bone as hydroxyapatite). Extracellular calcium is held in a very narrow range because small shifts cause major dysfunction.

| Form | % of total | Details |
|---|---|---|
| **Ionised (free)** | ~50% | Physiologically active; affected by pH (acidosis raises it, alkalosis lowers it) |
| **Protein-bound** | ~41% | Mainly albumin-bound; not active; reduced by hypoalbuminaemia |
| **Complexed** | ~9% | Bound to citrate, phosphate, sulphate, oxalate; not immediately active |

**Corrected calcium and normal values**

Low albumin makes total calcium look falsely low. Corrected Ca = measured total Ca + 0.02 x (40 minus albumin in g/L).

Normal total calcium 2.1 to 2.6 mmol/L. Normal ionised calcium 1.0 to 1.3 mmol/L.

Calcium functions include structural support in bone, muscle contraction (via troponin-C or calmodulin-MLCK), neurotransmitter release, the plateau phase of the cardiac action potential, coagulation cofactor activity, second-messenger signalling, stabilising membrane excitability, and driving hormone secretion by exocytosis.

## 3.3 Calcium Homeostasis: Overview

Serum calcium is set by three hormones acting on three organs (bone, kidney, intestine). Ionised calcium is sensed by the CaSR on parathyroid chief cells and thyroid C-cells, triggering reciprocal changes in PTH and calcitonin.

| Hormone | Source | Stimulus | Net effect on serum Ca |
|---|---|---|---|
| **PTH** | Parathyroid chief cells | Low Ca, low Mg, high phosphate | Raises calcium |
| **Calcitriol** | Kidney (PTH-stimulated 1a-hydroxylase) | Low Ca, low phosphate, PTH | Raises calcium |
| **Calcitonin** | Thyroid C-cells | Hypercalcaemia, gastrin | Lowers calcium |

## 3.4 Actions of Parathyroid Hormone

PTH acts on bone, kidney, and (indirectly) intestine to raise serum calcium and lower serum phosphate, all through PTH1R and cAMP.

### On bone

Acutely (minutes to hours) PTH drives rapid calcium efflux from bone surfaces (osteocytic osteolysis). Chronically it drives resorption: PTH binds PTH1R on osteoblasts (osteoclasts lack the receptor), osteoblasts release RANKL, RANKL activates osteoclast precursors, and activated osteoclasts secrete cathepsin K and acid phosphatases to dissolve matrix and release calcium and phosphate. Osteoprotegerin is a decoy that antagonises RANKL.

> **Intermittent versus continuous PTH**
>
> Intermittent (pulsatile) PTH is anabolic to bone, favouring osteoblasts. This is the basis of teriparatide (PTH 1-34) for osteoporosis.
>
> Continuous (sustained) PTH is catabolic, driving osteoclastic resorption, as in primary hyperparathyroidism (osteitis fibrosa cystica in severe disease).

### On the kidney

| Renal site | PTH action | Effect |
|---|---|---|
| **Proximal tubule** | Inhibits Na-phosphate co-transport (NaPi-2a) | Phosphaturia, lowers serum phosphate |
| **Distal tubule** | Stimulates calcium reabsorption via TRPV5 | Retains calcium, raises serum calcium |
| **Proximal tubule** | Up-regulates 1a-hydroxylase | More calcitriol, indirect gut calcium absorption |

### Net effect

PTH raises serum calcium, lowers serum phosphate (renal wasting dominates even though bone releases phosphate), raises calcitriol, and raises alkaline phosphatase with chronic elevation.

## 3.5 Actions of Calcitonin

Calcitonin is a 32-amino acid peptide from thyroid C-cells, released in hypercalcaemia and signalling through cAMP. It is the physiological brake on calcium, opposing PTH and calcitriol: it inhibits osteoclasts (less bone release), increases urinary calcium loss, and reduces gut absorption.

**Clinical note on calcitonin**

Calcitonin plays a minor role in adult calcium balance (thyroidectomy rarely causes hypercalcaemia). It matters most in fetal and neonatal life and in pregnancy and lactation, protecting the skeleton. Salmon calcitonin treats hypercalcaemia and Paget disease. Medullary thyroid carcinoma arises from C-cells and secretes calcitonin, a useful tumour marker.

## 3.6 Actions of Calcitriol (Active Vitamin D)

Vitamin D is activated in three steps: skin converts 7-dehydrocholesterol to cholecalciferol under UV-B; the liver adds a hydroxyl (25-hydroxylase) to make calcidiol; the kidney adds another (1a-hydroxylase, stimulated by PTH, low calcium, low phosphate) to make calcitriol, the active form. When calcium is high, 24-hydroxylase diverts the pathway to an inactive product.

| Target organ | Actions of calcitriol | Net effect |
|---|---|---|
| **Intestine (primary)** | Up calbindin, TRPV6, Ca-ATPase; phosphate absorption via NaPi | Absorbs calcium and phosphate |
| **Kidney** | Up calcium reabsorption in DCT | Raises serum calcium |
| **Bone** | Stimulates RANKL and osteoclasts; also osteoblast differentiation | Raises serum calcium and phosphate |
| **Parathyroid** | Down-regulates PTH gene transcription | Negative feedback |

Net effect of calcitriol: raises both serum calcium and serum phosphate, unlike PTH which lowers phosphate.

## 3.7 Disorders of Calcium Regulation

### Hypercalcaemia (serum Ca above 2.6 mmol/L)

| Cause | Mechanism | Lab clue |
|---|---|---|
| **Primary hyperparathyroidism** | Adenoma (85%), hyperplasia, carcinoma | High PTH, high Ca, low phosphate |
| **Malignancy** | PTHrP, osteolytic mets, calcitriol (lymphoma) | Low PTH, high Ca, high PTHrP |
| **Vitamin D toxicity** | Excess calcitriol or granulomatous disease | Low PTH, high calcitriol |
| **Milk-alkali syndrome** | Excess calcium and absorbable alkali | Metabolic alkalosis, normal/low PTH |

**Symptoms of hypercalcaemia: bones, stones, groans, psychic moans**

Bones: bone pain, fractures, osteitis fibrosa cystica. Stones: renal calculi, nephrocalcinosis. Groans: nausea, vomiting, constipation, pancreatitis. Psychic moans: depression, confusion, coma. Cardiac: shortened QT interval.

### Hypocalcaemia (serum Ca below 2.1 mmol/L)

| Cause | Mechanism / features |
|---|---|
| **Hypoparathyroidism** | Surgical (commonest), autoimmune, DiGeorge; tetany, prolonged QTc |
| **Vitamin D deficiency** | Poor diet/sunlight, malabsorption; rickets in children, osteomalacia in adults |
| **Pseudohypoparathyroidism** | PTH resistance (GNAS); high PTH; Albright osteodystrophy |
| **Hypomagnesaemia** | Mg needed for PTH secretion and action; refractory until Mg corrected |

**Clinical signs of hypocalcaemia**

Chvostek sign: tapping over the facial nerve triggers facial twitching. Trousseau sign: a BP cuff above systolic for 3 minutes provokes carpal spasm. Tetany: painful spontaneous cramps, from reduced membrane stabilisation lowering the sodium channel threshold. ECG: prolonged QTc with risk of Torsades de Pointes.

# 4 PANCREATIC ENDOCRINE PHYSIOLOGY

The pancreas is a mixed gland (about 15 cm, 60 to 100 g) lying retroperitoneally near the duodenum, portal vein, and superior mesenteric vessels. It is 98% exocrine and 2% endocrine (the islets). The exocrine pancreas secretes 1.5 to 2 L per day of alkaline, enzyme-rich juice (amylase, lipases, and protease zymogens such as trypsinogen) that neutralises gastric chyme and digests food. Secretin drives the bicarbonate-rich component and CCK the enzyme-rich component during the dominant intestinal phase.

## 4.1 Islets of Langerhans

The islets are 1 to 2% of pancreatic mass but receive around 20% of its blood flow, reflecting high activity. There are roughly 1 million islets.

| Cell | Proportion | Hormone | Function |
|---|---|---|---|
| **Beta** | 60 to 80% | Insulin | Lowers blood glucose; anabolic (storage) |
| **Alpha** | 15 to 20% | Glucagon | Raises blood glucose; catabolic (mobilisation) |
| **Delta** | 5 to 10% | Somatostatin | Inhibits both insulin and glucagon |
| **F (PP)** | 15 to 20% | Pancreatic polypeptide | Inhibits exocrine secretion; modulates motility |
| **Epsilon** | <1% | Ghrelin | Stimulates hunger |

## 4.2 Insulin: Structure, Secretion & Action

Insulin is a 51-amino acid polypeptide of an A chain (21) and B chain (30) joined by disulphide bonds. It comes from preproinsulin, then proinsulin; in the secretory granule C-peptide is cleaved off so insulin and C-peptide are secreted in equal amounts. C-peptide is measured clinically to assess endogenous secretion, because it is not present in injected insulin.

| Stimulus | Effect on insulin | Mechanism |
|---|---|---|
| **High blood glucose (primary)** | Potent stimulator | GLUT-2 uptake raises ATP/ADP, closes K-ATP channels, depolarisation, Ca influx, exocytosis |
| **Amino acids (arginine, leucine)** | Stimulator | Especially after a protein meal; synergises with glucose |
| **GIP and GLP-1 (incretins)** | Stimulator (glucose-dependent) | Gut hormones that amplify insulin; basis of GLP-1 agonists |
| **Parasympathetic (ACh)** | Stimulator | M3 receptors, IP3/calcium |
| **Sympathetic (alpha-2)** | Potent inhibitor | During stress; keeps glucose for brain and muscle |
| **Somatostatin** | Inhibitor | Paracrine from delta cells |

The insulin receptor is an alpha2-beta2 tyrosine kinase. Insulin binding autophosphorylates the beta subunits, which phosphorylate IRS proteins. IRS then drives the PI3K-Akt pathway (GLUT4 translocation, glycogen and protein synthesis) and the RAS-MAPK pathway (growth and gene expression).

| Tissue | Anabolic (fed-state) action of insulin |
|---|---|
| **Liver** | Up glycogen synthesis, glycolysis, lipogenesis; down gluconeogenesis, glycogenolysis, ketogenesis |
| **Skeletal muscle** | Up GLUT4 glucose uptake, glycogen and protein synthesis; down protein catabolism |
| **Adipose** | Up GLUT4 uptake and lipogenesis; down lipolysis |
| **All cells** | Up potassium uptake (basis of dextrose plus insulin for hyperkalaemia) |

## 4.3 Glucagon: Structure, Secretion & Action

Glucagon is a 29-amino acid peptide from alpha cells and the main counter-regulatory hormone to insulin, raising blood glucose during fasting and stress. Its secretion mirrors insulin: stimulated by hypoglycaemia, amino acids, sympathetic activity, and exercise; inhibited by high glucose, insulin, somatostatin, and free fatty acids.

| Tissue | Glucagon action | Consequence |
|---|---|---|
| **Liver** | Up glycogenolysis | Rapid glucose release |
| **Liver** | Up gluconeogenesis (PEPCK) | Sustained glucose production |
| **Liver** | Down glycolysis, up ketogenesis | Spares glucose; ketones as alternative fuel |
| **Adipose** | Up lipolysis | Provides substrate for the liver |

## 4.4 Counter-Regulatory Hormones

Several hormones oppose insulin to defend blood glucose during fasting and stress, which is central to managing diabetic hypoglycaemia.

| Hormone | Source | Anti-insulin mechanism | Importance |
|---|---|---|---|
| **Glucagon** | Alpha cells | Up hepatic glycogenolysis and gluconeogenesis, lipolysis | Primary, first-line defence |
| **Adrenaline** | Adrenal medulla | Up glycogenolysis and lipolysis; inhibits insulin | Critical when glucagon deficient |
| **Cortisol** | Adrenal cortex | Up gluconeogenesis, protein catabolism; reduces insulin sensitivity | Sustained hypoglycaemia (hours) |
| **Growth hormone** | Anterior pituitary | Up lipolysis and gluconeogenesis; reduces glucose uptake | Acts over hours |

# 5 ADRENAL PHYSIOLOGY & DISORDERS

The adrenal glands are paired retroperitoneal organs (4 to 5 g each) on the upper poles of the kidneys. The right is pyramidal and drains directly into the IVC (a surgical risk); the left is crescentic and drains via the left renal vein. Both are essential for life.

| Zone | % cortex | Hormone | Unique enzyme |
|---|---|---|---|
| **Zona glomerulosa (outer)** | 15% | Mineralocorticoids (aldosterone) | Aldosterone synthase (CYP11B2) |
| **Zona fasciculata (middle)** | 75% | Glucocorticoids (cortisol) | 17a-hydroxylase, 11b-hydroxylase |
| **Zona reticularis (inner)** | 10% | Androgens (DHEA) | 17,20-lyase |
| **Medulla (core)** | n/a | Catecholamines (adrenaline 80%) | PNMT (induced by cortisol) |

**Memory aid**

GFR going deeper: Glomerulosa = salt (goodies), Fasciculata = sugar (fat, cortisol), Reticularis = sex (androgens). The medulla, a modified sympathetic ganglion, makes catecholamines.

## 5.1 Glucocorticoids (Cortisol)

Cortisol provides about 95% of glucocorticoid activity and is made in the zona fasciculata from cholesterol. It is controlled by the HPA axis: hypothalamic CRH (pulsatile, peaking early morning) drives pituitary ACTH (from POMC), which binds MC2R on fasciculata cells to stimulate synthesis. Rising cortisol feeds back to suppress CRH and ACTH, but stress inputs override this feedback.

| System | Cortisol effect |
|---|---|
| **Carbohydrate** | Up gluconeogenesis and hepatic glycogen; down peripheral glucose uptake (steroid diabetes) |
| **Protein** | Up catabolism in muscle, skin, bone to supply amino acids |
| **Fat** | Up lipolysis; excess causes truncal obesity, buffalo hump, moon face, thin limbs |
| **Anti-inflammatory** | Stabilises lysosomes; inhibits phospholipase A2; fewer prostaglandins and cytokines |
| **Immune** | Lymphopaenia, eosinopaenia; neutrophilia (demargination) |
| **Permissive** | Needed for vascular tone, catecholamine sensitivity, and free water clearance |

### Cortisol disorders

| Disorder | Cause | Key features | Investigation |
|---|---|---|---|
| **Cushing syndrome** | Exogenous steroids (commonest), Cushing disease (pituitary ACTH), ectopic ACTH, adrenal tumour | Central obesity, moon face, striae, myopathy, hypertension, hyperglycaemia | 24h urinary cortisol; overnight dexamethasone suppression |
| **Addison disease (primary AI)** | Autoimmune (70%), TB, metastases | Fatigue, weight loss, postural hypotension, hyperpigmentation, low Na, high K | Low AM cortisol, high ACTH, failed Synacthen test |
| **Congenital adrenal hyperplasia** | 21-hydroxylase deficiency | Cortisol deficiency, androgen excess, salt-wasting crisis | Very high 17-OH-progesterone |

## 5.2 Mineralocorticoids (Aldosterone)

Aldosterone provides about 90% of mineralocorticoid activity and is made only in the zona glomerulosa. It is the main regulator of sodium balance, extracellular volume, and blood pressure. The strongest stimulus is angiotensin II through the RAAS; a rise in serum potassium is also very potent, while ACTH is only permissive.

Acting on principal cells of the distal tubule and collecting duct, aldosterone binds the mineralocorticoid receptor and increases apical ENaC channels, basolateral Na/K-ATPase, and apical ROMK channels. The result is sodium reabsorption, potassium secretion, and hydrogen ion secretion (urine acidified, blood alkalinised), with water following sodium to expand extracellular volume.

**Aldosterone disorders**

Primary hyperaldosteronism (Conn syndrome): adrenal adenoma or hyperplasia giving autonomous aldosterone, causing hypertension, hypokalaemia, metabolic alkalosis, and suppressed renin. It accounts for about 10% of hypertension; diagnosed by the aldosterone-to-renin ratio.

Deficiency (as in Addison): sodium and water loss with hyperkalaemia and dangerous cardiac toxicity (peaked T waves).

## 5.3 Adrenal Androgens & Catecholamines

The zona reticularis makes weak androgens (DHEA, DHEAS, androstenedione) that are converted peripherally to testosterone and, in fat, to oestrogens. In women the adrenal supplies most circulating androgens, important for libido and muscle.

The medulla is a modified sympathetic ganglion of chromaffin cells that make catecholamines (adrenaline 80%, noradrenaline 20%). Tyrosine is converted through L-DOPA, dopamine, and noradrenaline to adrenaline; PNMT (the final step) is induced by cortisol arriving from the cortex. Adrenergic receptors mediate the effects: alpha-1 (vasoconstriction), alpha-2 (reduced noradrenaline and insulin release), beta-1 (increased heart rate and contractility, renin), beta-2 (bronchodilation, glycogenolysis), and beta-3 (lipolysis).

**PHAEOCHROMOCYTOMA**

A catecholamine-secreting tumour of chromaffin cells (90% adrenal; extra-adrenal ones are paragangliomas). Classic triad: episodic hypertension, headache, sweating, with palpitations. Rule of 10s: 10% malignant, bilateral, extra-adrenal, and familial (MEN2, VHL, NF1).

Diagnosis: plasma metanephrines (most sensitive) or 24h urinary catecholamines. Treatment: alpha-blockade first (phenoxybenzamine), then beta-blockade, then surgery, to avoid an unopposed-alpha hypertensive crisis.

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### How to use this guide

Work through one section at a time. Read the prose for understanding, then use the tables and callout boxes for rapid revision and last-minute recall. Coloured boxes flag core principles, clinical pearls, drug targets, and emergencies. Every fact here has been rewritten in original wording from primary physiology teaching material.